

# Laplace Transforms

## Definitions and Notations from Prof. H4

### Laplace Transform

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$

### Common Laplace Transforms

1. Constant  $\mathcal{L}\{k\} = \frac{k}{s}$

2. Exponential  $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$

3. Trigonometric  $\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}$

$$\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2}$$

4. Power  $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$

### Properties

1. Linearity  $\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$

2. Shift  $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$

3. Derivative  $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$

$$\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0)$$

12. Compute the following Laplace Transforms:

a)  $\mathcal{L}\{1\}$

$\Rightarrow$  We use the constant rule:

$$\mathcal{L}\{k\} = \frac{k}{s}$$

$$\Rightarrow \mathcal{L}\{1\} = \frac{1}{s}$$

b)  $\mathcal{L}\{e^{-4t}\}$

$\Rightarrow$  We use the exponential rule:

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$\Rightarrow \mathcal{L}\{e^{-4t}\} = \frac{1}{s+4}$$

c)  $\mathcal{L}\{5u(t)\} = \mathcal{L}\{5 \cdot 1\} = \mathcal{L}\{5\}$

$\Rightarrow$  We use the constant rule:

$$\mathcal{L}\{k\} = \frac{k}{s}$$

$$\Rightarrow \mathcal{L}\{5\} = \frac{5}{s}$$

d)  $\mathcal{L}\{\cos(3t)\}$

$\Rightarrow$  We apply the Trig rule:

$$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}$$

$$\Rightarrow \mathcal{L}\{\cos(3t)\} = \frac{s}{s^2 + 3^2} = \frac{s}{s^2 + 9}$$

e)  $\mathcal{L}\{2\sin(4t)\}$

$\Rightarrow$  We use Linearity property and the Trig rule:

$\Rightarrow \mathcal{L}\{2\sin(4t)\} = 2\mathcal{L}\{\sin(4t)\}$

$\Rightarrow$  Trig rule  $\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2}$

$\Rightarrow \mathcal{L}\{\sin(4t)\} = \frac{4}{s^2 + 16}$

$\Rightarrow 2\mathcal{L}\{\sin(4t)\} = 2\left(\frac{4}{s^2 + 16}\right) = \frac{8}{s^2 + 16}$

f)  $\mathcal{L}\{t\}$

$\Rightarrow$  We use the Power rule:

$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$

$\Rightarrow \mathcal{L}\{t\} = \mathcal{L}\{t^1\} = \frac{1!}{s^{1+1}} = \frac{1}{s^2}$

[13]

Use the Linearity and Shift properties to compute the following:

a)  $\mathcal{L}\{4 - 3e^{-2t} + 2\sin(t)\}$

$\Rightarrow 4\mathcal{L}\{1\} - 3\mathcal{L}\{e^{-2t}\} + 2\mathcal{L}\{\sin(t)\}$

①  $\uparrow$   
Constant rule

②  $\uparrow$   
Exponential rule

③  $\uparrow$   
Trig rule

①  $\frac{4}{s}$

②  $-\frac{3}{s - (-2)} = -\frac{3}{s + 2}$

③  $\frac{2 \cdot 1}{s^2 + 1} = \frac{2}{s^2 + 1}$

$$\Rightarrow \mathcal{L}\{4 - 3e^{-2t} + 2\sin(t)\} = \frac{4}{s} - \frac{3}{s+2} + \frac{2}{s^2+1}$$

b)  $\mathcal{L}\{e^{-3t}\cos(2t)\}$

$\Rightarrow$  Shift: Multiplying by  $e^{-3t}$  shifts  $\cos(2t)$  in the  $s$ -domain by  $-3$ .

① Find the base transformation  $\mathcal{L}\{\cos(2t)\}$ !

$$\Rightarrow \mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}$$

$$\Rightarrow \mathcal{L}\{\cos(2t)\} = \frac{s}{s^2 + 2^2} = \frac{s}{s^2 + 4} \quad \checkmark$$

② Apply our shift of  $-3$  to get:  $F(s+3)$

Final Laplace transformation:

$$\Rightarrow \mathcal{L}\{e^{-3t}\cos(2t)\} = \frac{s+3}{(s+3)^2 + 4}$$

15) Solve the initial value problem by Laplace transform:

$$\dot{B}(t) + 2B(t) = 8, \quad B(0) = 1.$$

①  $\Rightarrow$  We use the Derivative rule:

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

②  $\Rightarrow$  We also use the Linearity property:

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$$

③  $\Rightarrow$  We finally use the constant rule:

$$\mathcal{L}\{k\} = \frac{k}{s}$$

Here we go:

$$\textcircled{1} \mathcal{L}\{B(t)\} = sB(s) - B(0) = sB(s) - 1$$

$$\textcircled{2} \mathcal{L}\{2B(t)\} = 2\mathcal{L}\{B(t)\} = 2B(s)$$

$$\textcircled{3} \mathcal{L}\{8\} = \frac{8}{s}$$

Put it all together:

$$sB(s) - 1 + 2B(s) = \frac{8}{s}$$

$\Rightarrow$  Determine both  $B(s)$  and  $B(t)$

$$sB(s) - 1 + 2B(s) = \frac{8}{s}$$

$$sB(s) + 2B(s) = \frac{8}{s} + 1$$

$$(s+2)B(s) = \frac{8}{s} + 1 = \frac{8+s}{s}$$

$$\Rightarrow B(s) = \frac{8+s}{s(s+2)}$$

$\Rightarrow$  To get our time domain function  $B(t)$ , we use Partial Fraction Decomp:

$$\frac{8+s}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2} = \frac{A(s+2) + Bs}{s(s+2)}$$

$$\text{Numerator} \Rightarrow 8+s = A(s+2) + Bs$$

Roots  $s=0: 8+0 = A(0+2) + 0$

$$8 = 2A$$

$$A = 4 \quad \checkmark$$

$$s = -2: \quad 8 - 2 = A(0) + B(-2)$$

$$6 = -2B$$

$$B = -3 \checkmark$$

$$\Rightarrow \text{Plug back in: } B(s) = \frac{A}{s} + \frac{B}{s+2} = \frac{4}{s} - \frac{3}{s+2}$$

$$\Rightarrow \text{Recognize } \frac{4}{s} \Rightarrow \text{Constant rule: } \mathcal{L}\{k\} = \frac{k}{s}$$

$$\Rightarrow \text{Recognize } \frac{-3}{s+2} \Rightarrow \text{Exponential rule + Linearity property}$$

$$\mathcal{L}\{af(t) \dots\} = aF(s) \dots$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$\Rightarrow B(t) = 4 - 3e^{-2t}$$

Double  
check:

$$\Rightarrow B(s) = \mathcal{L}\{4\} - 3\mathcal{L}\{e^{-2t}\}$$

↓

$$B(s) = \frac{4}{s} - \frac{3}{s+2} \quad \checkmark$$

20) The throughput response  $y(t)$  of an order-processing unit is modeled by

$$4\dot{y}(t) + y(t) = 6u(t), \quad y(0) = 0.$$

Use the Laplace transform to determine:

a)  $Y(s)$

We list our rules here:

① Derivative rule:

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

② Linearity Property:

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$$

③ Constant rule:

$$\mathcal{L}\{k\} = \frac{k}{s}$$

$\Rightarrow$  Start by transforming the first term of the diff eq.

$$\mathcal{L}\{4\dot{y}(t)\}$$

\* Apply ①, ②

$$\begin{aligned}\mathcal{L}\{4\dot{y}(t)\} &= 4[sY(s) - y(0)] \\ &= 4[sY(s)] \quad | \quad y(0) = 0\end{aligned}$$

$\Rightarrow$  Apply the transformation to the second term:

$$\mathcal{L}\{y(t)\} = Y(s)$$

$\Rightarrow$  Apply the transformation to the forcing term:

$$\begin{aligned}\mathcal{L}\{6u(t)\} &= \mathcal{L}\{6\} \quad | \quad \textcircled{3} \\ &= \frac{6}{s}\end{aligned}$$

Putting it all together, we get our Laplace transform:

$$4sY(s) + Y(s) = \frac{6}{s}$$

$$\Rightarrow (4s + 1)Y(s) = \frac{6}{s}$$

$$Y(s) = \frac{6}{s(4s + 1)}$$

b) a partial fraction decomp. of  $Y(s)$

$$Y(s) = \frac{6}{s(4s + 1)} = \frac{A}{s} + \frac{B}{4s + 1} = \frac{A(4s + 1) + Bs}{s(4s + 1)}$$

$$\Rightarrow \text{Numerator: } 6 = A(4s + 1) + Bs$$

$$\begin{array}{l|l} 4s + 1 = 0 & Bs = 0 \\ 4s = -1 & s = 0 \checkmark \\ s = -\frac{1}{4} & \end{array}$$

$$s = -\frac{1}{4}: \quad 6 = A(4(-\frac{1}{4}) + 1) + B(-\frac{1}{4})$$

$$6 = -\frac{1}{4}B$$

$$24 = -B$$

$$B = -24 \checkmark$$

$$s = 0: \quad 6 = A(0+1) + B(0)$$

$$A = 6 \quad \checkmark$$

$$Y(s) = \frac{A}{s} + \frac{B}{4s+1} = \frac{6}{s} - \frac{24}{4s+1}$$

c) find  $y(t)$

$\Rightarrow$  start by looking at the terms of our Laplace Transform

$$\frac{6}{s} \Rightarrow \text{Constant}$$

$$-\frac{24}{4s+1} = \frac{6}{s+1/4} \Rightarrow \text{Exponential + Linearity}$$

$$Y(s) = \mathcal{L}\{6\} - 6\mathcal{L}\{e^{-t/4}\}$$

$$y(t) = 6 - 6e^{-t/4}$$

d) the final value  $y(\infty)$

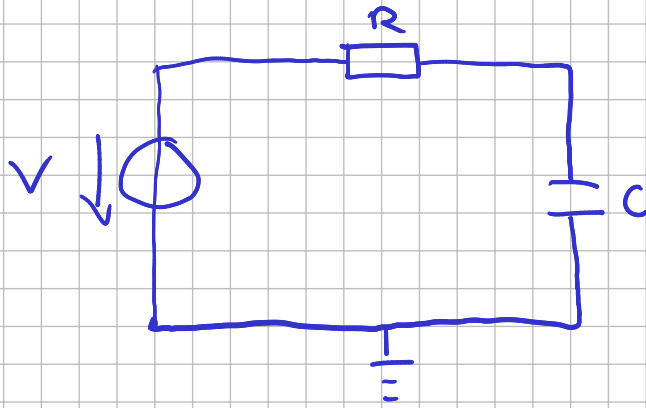
$$\Rightarrow \lim_{t \rightarrow \infty} 6 - 6e^{-t/4} = 6$$

e) the time constant of the system

$\Rightarrow$  To get the time constant, we look at how the system grows or decays:

$$e^{-t/4} \quad \tau = 4$$

RC circuit :



Capacitor at  $t = 0$   
is a short circuit

Capacitor at  $t = \infty$   
is an open circuit

Current through C decays  
at:

$$e^{-t/RC}$$

$$\tau = RC$$

$\tau$  is the time constant

$RC$

at  $5\tau$ , the capacitor  
99% open circuit.